

**DECCAN EDUCATION SOCIETY'S
DES PUNE UNIVERSITY, PUNE**
December-2025

School:	School of Engineering and Technology	Department:	Computer Engineering and Technology
Year:	Second Year	Program:	Bachelor of Technology – CSE/ECE-AIML
Course Code:	ETCS300	Course Name:	Data Structures – Regular
Semester:	I	Total Pages:	2
Max. Marks:	50	Max. Time:	2 Hours
Pattern:	2025	PRN/Seat No.:	

Instructions to Candidates

1. Figures to the right indicate full marks.
2. Read all questions and instructions carefully before attempting.
3. Use of cell phone and smart watch is prohibited in the examination hall.
4. Attempt ANY TWO sub-questions from each question.

Q. NO.	Question Statement	Marks
Q.1	Attempt any TWO questions from the following	10
a)	Use the Frequency (Step) Count Method to determine the time complexity of the following code. Construct a step count table showing each statement, its frequency, and total steps. <pre>for (int i = 1; i <= n; i++) { for (int j = 1; j <= i; j++) { x = x + 1; } }</pre>	5
b)	For the Binary Search code given below, determine the Best Case, Average Case, and Worst Case time complexity . Express each using asymptotic notation and justify your answer. <pre>int binarySearch(int arr[], int n, int key) { int low = 0, high = n - 1; while (low <= high) { int mid = (low + high) / 2; if (arr[mid] == key) return mid; else if (arr[mid] < key) low = mid + 1; else high = mid - 1; } return -1; }</pre>	5
c)	Write the C++ code snippet for Insertion Sort . Trace all passes for the array [8, 3, 5, 1, 6] showing the array state after each pass. State its best case and worst case time complexity.	5
Q.2	Attempt any TWO questions from the following	10
a)	For the Binary Tree given below, write the Inorder, Preorder, and Postorder traversals. Show the complete traversal sequence for each. Also mention which traversal produces a sorted sequence for a BST and why. <pre>10 / \ 5 20 / \ / \ 3 7 15 25</pre>	5

b)	Construct an AVL Tree by inserting the following keys one by one: 30, 20, 40, 10, 25, 35, 50, 5 . After each insertion that causes imbalance, identify the rotation type (LL / RR / LR / RL) and draw the tree after rebalancing.	5
c)	What is a Threaded Binary Tree ? Distinguish between single-threaded and double-threaded binary trees with diagrams. Write an algorithm to perform Inorder traversal on a right-threaded binary tree without using a stack or recursion.	5
Q.3 Attempt any TWO questions from the following		10
a)	For a Binary Search Tree , write C++ functions for Insert , Search , and Delete operations. Demonstrate deletion of a node with two children using the in-order successor. Trace the deletion of key 20 from the BST formed by: 30, 20, 40, 10, 25.	5
b)	Implement a Min-Heap using an array. Write functions for Insert and Extract-Min with heapify. Show step-by-step insertion of keys [15, 10, 8, 12, 5, 3] into an initially empty min-heap, drawing the heap after each insertion.	5
c)	Compare B-Tree and B+ Tree with respect to structure, data storage, and range query performance. For a B+ Tree of order 3, show the insertion of keys: 5, 10, 15, 20, 25, 30 , drawing the tree after each split.	5

Q. NO.	Question Statement	Marks
Q.4	Attempt any TWO questions from the following	10
a)	Apply Dijkstra's Single Source Shortest Path algorithm on the weighted graph below starting from vertex A. Show the distance table after each step and state the shortest path to each vertex. Vertices: {A, B, C, D, E, F} Edges (undirected): A-B(4), A-C(2), B-C(1), B-D(5), C-D(8), C-E(10), D-F(2), E-F(3)	5
b)	Apply Kruskal's Algorithm on the same graph above to find the Minimum Spanning Tree. Show the Union-Find steps, the edges selected in order, and the total MST cost. How does Kruskal's differ from Prim's algorithm?	5
c)	Perform Topological Sort on the DAG given below using DFS-based approach (stack method). Also check whether the graph is a valid DAG. State one real-world application of topological ordering. Edges (directed): 5→2, 5→0, 4→0, 4→1, 2→3, 3→1 Vertices: {0, 1, 2, 3, 4, 5}	5
Q.5	Attempt any TWO questions from the following	10
a)	A hash table of size 11 uses the hash function $h(k) = k \bmod 11$. Insert the following keys: 22, 33, 44, 55, 66, 77, 88 using: (i) Linear Probing (ii) Quadratic Probing (iii) Chaining. Show the final hash table for each method and compare the number of collisions.	5
b)	Explain Double Hashing as a collision resolution strategy. Using $h_1(k) = k \bmod 7$ and $h_2(k) = 5 - (k \bmod 5)$, insert keys [18, 41, 22, 44, 59, 32, 31, 73] into a hash table of size 7. Show the probe sequence for each key and the final table.	5
c)	(i) Using the Java Collections Framework, write a Java program that uses HashMap to store student roll numbers (key) and their marks (value). Perform: insert 5 students, update one student's marks, display all entries sorted by roll number using TreeMap. (ii) What is a Skip List? Draw a skip list for keys [3, 7, 12, 19, 26] with 3 levels and explain the search operation for key 19.	5

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All the Best!